

VIKALP THUKRAL

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EDUCATION

UCLA Anderson School of Management

Los Angeles, CA

Master of Financial Engineering

Dec 2025

- Honors: 2025 IAQF Competition Winner – Team Geeks for Greeks (Top 4/30 teams) – [Read the paper](#); GRE Quant: 167/170; GPA: 3.713

• Relevant Coursework: Advanced Computational Methods, ML & AI, Financial Modeling, Empirical Methods, Stocal, Prob & Stats

Punjab Engineering College

Chandigarh, India

B.Tech, Electronics and Communication (GPA: 3.75)

July 2021

- Relevant Coursework: Linear Algebra, Statistics, Data Structures & Algorithms, Object-Oriented Programming (C++)

SKILLS & CERTIFICATIONS

Programming: Python (Pandas, NumPy, SciPy, scikit-learn, FastAPI), C++, SQL, JavaScript (React), VBA

AI / ML & Agentic Development: Quant Toolkit Development with Agentic AI (Claude Code), LLM API Integration (Gemini, Anthropic), NLP / FinBERT, Gradient Boosting, Random Forest, XGBoost, Ridge / Lasso Regression, Prompt Engineering, REST API Design

Quantitative & Financial Analysis: Value at Risk (VaR), Expected Shortfall, GJR-GARCH, Stress Testing, Backtesting (Kupiec/Christoffersen), Derivatives Pricing, Monte Carlo, Time-Series Methods

Data Analysis & Big Data: PySpark, Hadoop, Apache Hive, Snowflake, Tableau, Power BI, Advanced Excel (Lookups, Pivots)

Certifications: CFA Level II Candidate, Machine Learning in Finance

PROFESSIONAL EXPERIENCE

UCLA Anderson School of Management

Los Angeles, CA

Quant Research Analyst

Feb 2026 – Present

- Develop Python-based quant research tooling and risk analytics modules in support of faculty research, leveraging agentic AI (Claude Code) to accelerate prototyping, calibration, and backtesting of derivative pricing and risk models across multiple asset classes.

• Conduct alpha research on the WorldQuant BRAIN platform; building, simulating, and submitting systematic equity alphas using its operator framework.

DoubleLine Capital — Applied Finance Project (UCLA MFE)

Los Angeles, CA

Quantitative Research Project

Sep 2025 – Dec 2025

- Designed an NLP sentiment pipeline using FinBERT to extract systematic signals from Fed and macro communications, generating quantitative features for credit and rates research.
- Engineered a Python ETL workflow to ingest, clean, and tokenize unstructured text data; packaged model outputs into reproducible datasets handed off to the firm's quant research team.

Link Capital LLC

Los Angeles, CA

Associate – Quantitative Risk Analytics

May 2025 – Nov 2025

- Developed enhancements to liquidity risk and portfolio monitoring tools using Python and automated SQL workflows, strengthening day-to-day risk infrastructure for the credit portfolio.
- Built and maintained risk reports and Tableau dashboards translating exposure, concentration, and policy-adherence metrics into actionable views for senior management.

American Express

Gurugram, India

Associate – Data Analytics & Product Management

Jan 2023 – June 2024

- Engineered ETL pipelines (PySpark, Hive, SQL) on 200M+ records for the customer data warehouse, ensuring data integrity and regulatory compliance.
- Built Python anomaly detection & reporting workflows; replaced legacy code paths to cut runtime & maintain strict policy adherence.

Associate – Product Data Analytics

Aug 2021 – Dec 2022

- Led data engineering integrations for regulatory frameworks; built executive Tableau dashboards monitoring KPIs and firm-wide compliance risks.

AGENTIC-AI QUANT TOOLKIT — PROJECTS | github.com/Weculp | vikalpthukral.com

Volatility Surface Explorer [[Live](#) · [GitHub](#)] **Tech:** Plotly.js, FastAPI, yfinance, curl_cffi, Docker, Fly.io

- Interactive 3D implied-volatility surface for any US-listed equity with switchable Z-axis metrics (IV, Total Variance, Volume, OI), bilinear resampling + Gaussian smoothing, ATM term-structure and skew mini-charts, and PNG/CSV export.

Risk Analytics Dashboard [[Live](#) · [GitHub](#)] **Tech:** React, Recharts, FastAPI, NumPy, SciPy, arch, Fly.io

- Institutional-grade portfolio risk dashboard: VaR (Historical, Parametric, Cornish-Fisher, GJR-GARCH-FHS, Bootstrap CI) across 1d–1y horizons, Expected Shortfall, Euler-component VaR, and 14 historical stress scenarios (Black Monday → SVB).

Commodities Dashboard [[Live](#) · [GitHub](#)] **Tech:** React, Plotly.js, FastAPI, arch, Google Gemini API, Fly.io

- Real-time futures analytics across 19 commodities × 5 sectors: term-structure (contango/backwardation), realized vol (10/21/63d, EWMA, GJR-GARCH), ETF-proxy implied vol, 10-yr seasonality heatmap, calendar-spread z-scores.

INTERESTS

WorldQuant Alpha Research, Competitive Gaming (Connect 4 ELO: 1950+), Creative Arts (Sketching, Cooking), Badminton, Fintech & AI Trends